Python Programming Assignment

UG IIM Students | Loops, Number Logic, Series & Problem-Solving

Learning objectives: Practice for and while loops, accumulators, conditions, number manipulation, series, input validation, and loop tracing.

Instructions

- Use loops and conditions to solve each problem; avoid built-in shortcuts that bypass the intended logic.
- Where specified, solve using both for and while loops.
- Test edge cases and include sample outputs with your submission.
- Nested-loop pattern questions are separated at the end as preview questions.

Section A — Beginner

Q1. Print numbers from 1 to N

Accept an integer N and print all numbers from 1 to N. Solve using both for and while loops.

Sample / test: Input: 5 | Output: 1 2 3 4 5

Q2. Multiplication table of 7

Print the multiplication table of 7 from 1 to 10. Then modify the program to print the table of any number entered by the user.

Sample / test: Example: 7 × 3 = 21

Q3. Sum of the first N natural numbers

Calculate 1 + 2 + 3 + ... + N using a loop. Do not use the formula N(N+1)/2. Solve using for and while.

Sample / test: Input: 5 | Output: 15

Q4. Sum of squares

Calculate $1^2 + 2^2 + 3^2 + ... + N^2$ using a loop.

Sample / test: Input: 4 | Output: 30

Q5. Sum of even and odd numbers

From 1 to N, calculate the sum of even numbers and the sum of odd numbers separately.

Sample / test: Input: 6 | Output: Even sum = 12; Odd sum = 9

Q6. Factorial of a number

Calculate N! for a non-negative integer. Do not use math.factorial(). Handle 0! = 1. Solve using for and while.

Sample / test: Input: 5 | Output: 120

Section B — Beginner to Intermediate

Q7. Fibonacci series

Print the first N terms of the Fibonacci series, starting with 0 and 1. Solve using for and while.

Sample / test: Input: 7 | Output: 0 1 1 2 3 5 8

Q8. Sum of cubes

Calculate $1^3 + 2^3 + 3^3 + \ldots + N^3$ using a loop.

Sample / test: Input: 3 | Output: 36

Q9. Generate a squares series

Print 1, 4, 9, 16, ..., N^2 and calculate the sum.

Sample / test: Input: 4 | Output: Series = 1 4 9 16; Sum = 30

Q10. Arithmetic progression

Accept the first term, common difference, and number of terms. Print the series and its sum.

Sample / test: First term = 2, difference = 3, terms = 5 | Output: 2 5 8 11 14; Sum = 40

Q11. Count and sum digits

Count the digits of an integer and calculate their sum without converting the number to a string. Handle zero and negative inputs.

Sample / test: Input: 5832 | Output: Digits = 4; Sum = 18

Q12. Reverse a number

Reverse the digits of a non-negative integer using arithmetic operations and a while loop. Do not convert it to a string.

Sample / test: Input: 12340 | Output: 4321

Section C — Intermediate

Q13. Number palindrome

Check whether an integer reads the same forward and backward. Do not convert it to a string. State how negative numbers are handled.

Sample / test: Input: 12321 | Output: Palindrome

Q14. Prime number checker

Determine whether an integer is prime using a loop to test divisibility. Handle numbers less than 2.

Sample / test: Input: 17 | Output: Prime; Input: 21 | Output: Not prime

Q15. Print all primes up to N

Print every prime number from 2 through N. Use nested loops to test each candidate.

Sample / test: Input: 10 | Output: 2 3 5 7

Q16. Sum of factorials

Calculate 1! + 2! + 3! + … + N!. Reuse the previous factorial result instead of recalculating from scratch.

Sample / test: Input: 4 | Output: 33

Q17. Sum of reciprocal series

Calculate S = 1 + 1/2 + 1/3 + … + 1/N. Display the result to four decimal places.

Sample / test: Input: 4 | Output: 2.0833

Q18. Geometric progression

Accept the first term, common ratio, and number of terms. Print the sequence and its sum.

Sample / test: First term = 2, ratio = 2, terms = 5 | Output: 2 4 8 16 32; Sum = 62

Section D — Advanced

Q26. Armstrong number

Check whether a number equals the sum of its digits raised to the power of the number of digits.

Sample / test: Input: 153 | Output: Armstrong number

Q27. Greatest Common Divisor (GCD)

Accept two integers and calculate their GCD using a loop and the Euclidean algorithm. Do not use math.gcd().

Sample / test: Input: 48, 18 | Output: 6

Q28. Perfect number

Determine whether a positive integer equals the sum of its proper positive divisors, excluding itself.

Sample / test: Input: 28 | Output: Perfect number

Q29. Alternating series

Print and calculate S = 1 − 2 + 3 − 4 + 5 − … for N terms.

Sample / test: Input: 5 | Output: 3

Q30. Menu-driven calculator

Build a calculator that repeatedly displays options for addition, subtraction, multiplication, division, and exit. Continue until the user chooses exit. Validate choices and prevent division by zero.

Sample / test: Use a while loop to repeat the menu.

Section E — Trick questions

Q31. String palindrome

Check whether a phrase is a palindrome without slicing ([::-1]) or reversed(). Ignore spaces and letter case. Then decide and document how punctuation should be handled.

Sample / test: Input: Never odd or even | Output: Palindrome

Q32. Is this an Indian mobile-number format?

Accept the number as a string. Consider a 10-digit Indian mobile number, optional +91 or 91 country code, and optional spaces. Normalize permitted separators before checking. Explain why format validation cannot prove that a number is active or assigned.

Sample / test: Examples: +919876543210 → Valid format; 1234567890 → Invalid format

Q33. Predict the output: for-else

Predict the output and explain what changes if break is executed inside the loop:

```
for i in range(1, 5):
  print(i, end=' ')
else:
  print('Done')
```

Sample / test: Explain when a loop's else block executes.

Q34. Count digits without string conversion

Count the digits of an integer using a while loop. Test 1234, 7, 0, and a negative number. Handle zero explicitly.

Sample / test: Input: 0 | Output: 1 digit

Q35. Infinite-loop trap

Predict what happens and correct this code:

```
i = 1
while i <= 5:
  print(i)
  i = i - 1
```

Sample / test: Explain why the condition never becomes false.

Q36. Nested-loop output

Predict the exact output:

```
for i in range(1, 4):
  for j in range(1, i + 1):
    print(j, end='')
  print()
```

Then modify it to print 1 / 22 / 333.

Sample / test: Explain the role of each loop.

Section F — Preview: Nested Loops and Pattern Printing

These questions are for preview and classroom discussion. Attempt them after completing the core assignment. Use nested loops where appropriate. Expected outputs below use N = 5 unless stated otherwise.

P1. Star triangle

Print a left-aligned triangle of stars for N rows.

Expected output:

```
*
* *
* * *
* * * *
* * * * *
```

P2. Inverted star triangle

Print N stars in the first row, decreasing by one star per row.

Expected output:

```
* * * * *
* * * *
* * *
* *
*
```

P3. Increasing number triangle

Print numbers from 1 to the current row number.

Expected output:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

P4. Repeated number triangle

Repeat the row number as many times as its value.

Expected output:

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

P5. Floyd's triangle

Print consecutive integers continuously across rows.

Expected output:

```
1
2 3
4 5 6
7 8 9 10
11 12 13 14 15
```

P6. Centered star pyramid

Print a centered pyramid. Each row has leading spaces and an odd number of stars.

Expected output:

```
    *
   ***
  *****
 *******
*********
```

P7. Centered number pyramid

Print increasing digits up to the row number, then decreasing digits back to 1.

Expected output:

```
    1
   121
  12321
 1234321
123454321
```

P8. Diamond pattern

Combine an increasing pyramid and a decreasing inverted pyramid without duplicating the middle row.

Expected output:

```
  *
 ***
*****
 ***
  *
```